

Laboration 1: Geostrophic adjustment

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1 Aim

The purpose of this lab is to understand what controls the evolution of an initial disturbance of a fluid at rest on a rotating plane, more commonly known as geostrophic adjustment. To study this phenomena, you will use a simplified ocean model based on the linear shallow water equations.

$$\frac{\partial u}{\partial t} - f_0 v = -g \frac{\partial \eta}{\partial x} \quad (1)$$

$$\frac{\partial v}{\partial t} + f_0 u = -g \frac{\partial \eta}{\partial y} \quad (2)$$

$$\frac{\partial \eta}{\partial t} + D_0 \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = 0 \quad (3)$$

2 Model setup

The model, we are going to use, is based on the equations above. Either you can use the model you programmed during the Numerical Methods course (generally not recommended since it complicates discussions regarding the code), or you can use the model given on Athena (recommended). For each laboration and exercise you need to program new parts in the model to study different physical processes, waves and circulation in the ocean and atmosphere. The aim is always to compare your simulated solutions with

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the analytical solutions that you derived during the lectures.

In this laboratory you are supposed to study the geostrophic adjustment for an initial disturbance consisting of a discontinuity in η that is symmetric in y .

Before starting with the experiments, the following should be programmed in the code:

- Start by programming the Coriolis force in the model. Since we are going to study a disturbance on a rotating plane we need the Coriolis force in our model.
 - Remember that Coriolis needs to be implemented on all time steps (Euler forward and leapfrog)
 - Remember that we are using a C-grid, thus v and u are not on the same points.
 - Remember that periodic boundaries will affect the computation of Coriolis.
- Implement a disturbance in h , which is described by a step function in East - West. A good practice is to program the initial condition as a CASE or LOGICAL to be able to turn it on or off for other exercises.
- The code already describes a Gaussian disturbance of the sea surface height as an initial condition, **do not** take this away, instead make this as one of the options for the initial condition!
- Use periodic boundaries in north and south.
- Use open boundaries in east and west. Make sure that the sponge (S in the code) is large enough.
- Use parameters that are similar to the world ocean when running the experiments (although, sometimes values need to be unrealistic in order to ensure numerical stability). **But remember to make sure that you are resolving the Rossby radius, and that the domain is large enough to get a good relaxation. If your results differ too much from the analytical solution you maybe need to re-think the settings.**

$$L_x \sim 6 \cdot 10^6 m$$

$$L_y \sim 4.5 \cdot 10^6 m$$

$$D_0 \sim 4000 m$$

$$g = 9.81 m s^{-2}$$

$$f_0 = 10^{-4} s^{-1}$$

- Set fixed N_X and N_Y
- Let $\Delta x, \Delta y$ be determined by L_X, L_Y and N_X, N_Y
- Let Δt be determined by a Courant number, phase speed, and $\max(\Delta x, \Delta y)$.
- Let N_T be determined by $T_{\max}$, which is the simulation period
- Construct the main time loop so that the model **does not** store the fields every time step!!!

3 Experiments

Experiment 1 - Sea surface elevation and velocity

In this part you are supposed to discuss the final solution for the sea surface elevation η and the velocity v .

- Run the model until the system only varies very little in time (steady state).
- Tweak the parameters. Think about which parameters are most important.
- Answering the question sheet will help you to
 - verify your model results with the results you obtained analytically,
 - discuss the results from tweaking parameters.

Experiment 2 - Energetics

In this exercise you are supposed to study the different energetics of the system. The system can be considered to be in steady state when the energy of the system changes very little with time.

- Program the kinetic energy and the available potential energy in your Shallow Water Model.
- Run the model with two different depths; one shallow $D_0 \sim 500\text{m}$ and one deep $D_0 \sim 10000\text{m}$.
- Run the model with two different f .
- The question sheet will help you to
 - Study how the energies of the system (kinetic, potential, and total) vary in time for the two cases with different depth and with different coriolis.
 - Give a physical explanation of the results.
 - How do the results change? Try to explain the changes and compare with theory.

Good luck!